NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR304089
Lead Applicant	Mr Andrew Eaton
Research Title	The clinical and cost-effectiveness of mandibular advancement devices in managing severe obstructive sleep apnoea.
Plain English Summary	**Research Aims** To explore the effect of mandibular advancement therapy on the physical, mental, social and emotional health and wellbeing of people living with the severe obstructive sleep apnoea (OSA). **Research Background** OSA affects breathing during sleep, in which there is repeated partial or complete obstruction of their airway, resulting in dangerously low levels of oxygen in the body. This obstruction to breathing can significantly impact on the person's quality of sleep, with sleep deprivation resulting in an increased risk of road traffic accidents, memory loss and mood disturbance. The lack of oxygen has been shown to increase the risk of developing high blood pressure, heart attack and stroke. The accepted treatment for OSA is continuous positive airway pressure (CPAP), in which the airway is prevented from collapsing by a machine which blows air through a tight-fitting face mask. However, many people struggle to successfully use this machine for several reasons, among which the mask can feel claustrophobic the machine is noisy, and the air pressure is uncomfortable to the eyes and causes bloating of the stomach. A recognised alternative to CPAP is a mandibular advancement device (MAD), these resemble a gum shield and when worn at night, they hold the lower jaw and tongue forward, preventing the airway from collapsing. Importantly, patient's report greater preference for an MAD in place of CPAP. The current research is much needed, given the low patient acceptance of CPAP, leaving them vulnerable to a significant healthcare burden and the high patient expressed preference for MAD treatment. As yet, the benefits of MAD treatment in patients with severe OSA need to be demonstrated in order to offer these as an alternative option for care. **Design and Methods** I plan to conduct a clinical trial involving patients with severe OSA, designed to compare CPAP and MAD treatments looking at changes in their breathing, blood pressure and heart function. Patients will also complete questionnaires designed to look at their general health and mood status. One of these questionnaires is concerned with helping to calculate the costs of providing CPAP and MAD treatments and therefore help to determine their value-for-money. In total, the study will require three visits to the hospital over a six-month period. I also plan to invite these patients to take part in interviews, designed to explore their views on the two treatments and help identify what they feel are key factors for us to consider in helping to improve the treatments. **Public involvement and engagement** Prior to committing to this application, I undertook clinical observation in OSA-specialist clinics, and I spoke with patients with suspected-OSA. I engaged with the British Society of Dental Sleep Medicine and patient charities, OSA Alliance and the OSA Partnership Group. Patients articulated the need to improve their quality of life as they experienced challenges such as fatigue, tiredness, loss of concentration and irritability. Equally, patients described the need to identify symptoms and to improve individual treatment modalities to successfully improving functional aspects of living with OSA. Because of this engagement I was able to form a group of four patient / expert users who were prepared to support me with setting up this project. **Dissemination**

	The product of this research will be shared by the appropriate format in scientific conferences and publications in peer-reviewed journals. It will be shared with stakeholder networks and support groups, both online and via engagement activities to exchange knowledge and raise awareness of my work. Potentially my research could provide proven benefit to patients with this severely debilitating breathing disorder and cost-effectiveness in its approach to care.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
Barts Health NHS Trust

Partner organisation	Queen Mary University of London

Research Title	The clinical and cost-effectiveness of mandibular advancement devices in managing severe obstructive sleep apnoea.

Proposed WTE Please select which WTE option you would prefer
0.60

Duration (months) The duration for your award is based on your chosen WTE
60

Proposed start date if grant awarded	01/09/2024
Research Type	Pilot study

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2022	01/2023	Module Certificate	Advanced Research Communication	United Kingdom	University of the West of England, Bristol	Pass
01/2022	01/2023	Pre-Doctoral Bridging Programme	Health Education England / National Institute Health Research, Pre-Doctoral Bridging Programme	United Kingdom	Kings College London	Pass
01/2020	06/2020	Professional Qualification	Accredited Exercise Scientist	United Kingdom	Perioperative Cardiopulmonary Exercise Testing Society	Pass
04/2018	05/2018	Professional Qualification	Registered Polysomnographic Technologist	United Kingdom	Board of Registered Polysomnographic Technologists	Pass
09/2012	06/2014	MSc	Clinical & Experimental Respiratory Medicine	United Kingdom (England)	University College London	Pass
06/2005	06/2007	Professional Qualification	Association for Respiratory Technology & Physiology – Part 1 and 2 Exams		Association for Respiratory Technology & Physiology	Pass
09/2003	06/2007	BSc(Hons)	Clinical Physiology - Respiratory	United Kingdom (England)	Anglia Ruskin University	2:2

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
01/2019	-	Clinical Scientist	Barts Health NHS Trust	10
11/2016	01/2019	Clinical Scientist	Royal National Ear Nose and Throat Hospital	50
08/2009	11/2016	Consultant Research Physiologist	Respiratory Clinical Trials Ltd.	100
06/2008	08/2009	Senior Clinical Physiogist	Royal Brompton & Harefield NHS Foundation Trust	30
01/2008	06/2008	Locum Physiologist	Locum	0
09/2000	06/2007	Clinical Physiologist	Royal Papworth Hospital NHS Foundation Trust	10

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
HEE ICA PhD Bridging Programme	I was awarded a place as a fellow on HEE's ICA Bridging scheme, hosted by King's College London. This programme involved mentorship from within the HEI and the NHS, and included a taught module (PhD Funding Proposal Development). It aimed	5000	Higher Education England	12	01/01/2022	31/12/2023

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	to support me in developing a competitive DCAF application.					
Delmedica Investments Development Grant	Research Funding Award & Travel Grant MSc in Clinical and Experimental Respiratory Medicine, , 20% WTE was research activities.	10000	Delmedica Investments	24	01/09/2023	01/07/2014
Philip Morris International	A research and travel grant to CSEM, Neuchâtel for the purposes of prototype development.	1900	Philip Morris International	1	01/03/2011	01/04/2011

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
(2021) ‘Exhaled Breath Temperature Predicts Maximal Exercise Performance In A Population Of Males With COPD.’,
(2021) ‘Minute-Ventilation Variability During Cardiopulmonary Exercise Test Is Higher In Idiopathic Dyspnoea Subjects Compared To Subjects With Breathing Pattern Disorder’, *Research Square*.

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/07/2022	Research Design Service	Patient Public Involvement Engagement, Amount Received £350	PPI meeting leader
01/01/2022	Kings College London	PhD Funding Proposal Development in Healthcare Research module, Amount Received £2780	Pre Doctoral student
01/05/2005	International Lung Sounds Association	Best spoken abstract - young researcher category, 2011 conference.	award winner

ORCID iD 0000-0003-3430-2907

3. Applicant Research Background

Professional Background	Other Registered Health or Care Professional

Profession (Other Registered Health or Care Professional)
Healthcare Scientist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
CS19994 - HCPC

Please describe your research career to date
Throughout my career, I have aimed to develop my clinical and research skills in parallel and have been fortunate to participate in some inspiring clinical studies, aimed at improving patient care. This stimulated my ambition to become a research leader in sleep medicine. I continually engage in research development, attending conferences, presenting research, collaborating as an external reviewer and editorial board member. Furthermore, I have built relationships with multidisciplinary researchers and patients' groups. However, my formal research training and experience have come from the following sources: **MSc in Clinical and Experimental Respiratory Medicine;** In 2014, I completed this industry-funded degree with taught modules on research ethics and governance, quantitative, qualitative and systematic review methods and translational clinical science. For the dissertation, I undertook a parallel design research study that examined the sensitivity of an exhaled breath thermometer device in a population of subjects with established Chronic Obstructive Pulmonary Disease for which I was awarded a distinction grade. **Clinical Trials Delivery:** From 2009 to 2016 I was a Consultant Physiologist at *Respiratory Clinical Trials Ltd* (RCT). My role included study delivery and management; including, site initiation, subject consent through to completion and the dissemination of findings. Whilst at RCT my research interest focussed on the measurement of respiratory lung sounds. I worked with an American company (*Stethographics Inc*) that had produced a device for recording, displaying and identifying disease through acoustic modelling. I made more than 9000 recordings of trial subjects. The data collected was novel and exploratory. My work in lung sounds became developmental and with the collaboration with *Swiss Centre for Electronics and Microtechnology*, we designed, fabricated, and trialled a prototype wearable device. In 2014, I began acting as the Sub-Investigator on three lung sound studies. I was responsible for all aspects of these studies. It was the first opportunity I had to work as an autonomous researcher. I set up the trial, submitted an ethics committee application, conducted day-to-day activities, recruited volunteers, and performed the outcome measures. I received training from sponsors in Site File and Trial Master File management, Study Initiation, Quality Assurance and Protocol Deviation procedures. In 2016, as a Clinical Scientist within the NHS, I was the research coordinator for two clinical trials. The first trial investigated the safety and efficacy of an investigational medicinal product for the treatment of insomnia in Alzheimer's and the second investigated sleep disturbances Priapism. I am currently involved in the delivery of a multi-centre Phase 2 trial, examining chest and abdominal wall movement in chronic obstructive pulmonary disease using a novel technology. **NIHR courses:** To broaden my underpinning knowledge, I completed the following courses: Becoming Research

Active; GCP; Questionnaire Delivery; Qualitative Evaluation; Lay Statement Writing; Increasing Participation of Ethnic Minorities in Health Research; Equality, Diversity and Inclusion and Patient Involvement and Engagement. To improve my research leadership skills, I completed PI Essentials.

I have successfully used RDS funds for PPIE with four patient experts. Their input so far has been to review the lay statement, to understand the research question and to suggest questionnaires be in both paper and electronic format.

HEE Pre-Doctoral Clinical Research Fellowship: I found this invaluable, it allowed me to develop knowledge of research, evaluation, and methodology. To better understand the processes required for preparing a doctoral application. Being exposed to a research environment, I built on my understanding of the four pillars of Advanced Clinical Practice. Specifically, in preparation for the foreseen training plan, I undertook introductory courses in health economics and Dental Sleep Medicine. This training was strengthened by courses in utilising group data with NVivo and search strategies. The output of the course was a detailed research proposal in preparation for this application. It improved the structure and help to focus my argument so that they were insightful and impactful. The robustness of my proposal was tested in a mock panel interview, providing a critical awareness of the importance of communicating potential science to different audiences.

Applied Research: I completed this post-graduate module of further research training. I learnt critical reflection skills and acquired knowledge on a range of strategies which can be used to perform rigorous PhD research. My proposals were critiqued by my peers and tested by oral presentation.

Lecturing: I use current research as the foundation of my materials and lecturing as a catalyst and platform to improve my presentation and delivery skills, ensuring information is impactful.

Scientific Outputs: Despite a very busy clinical workload I have managed to produce the following academic outputs: two abstracts, three conference presentations and two written publications. In 2005, two abstracts on predicting exercise tolerance were presented by a colleague at the American Thoracic Society Conference. In 2011 at the International Lung Sounds conference, I presented data about a prototype wearable technology for the transdermal capture of cardio-respiratory signals. In 2018, I presented pilot study data at the BACO ENT Conference detailing the impact of a non-invasive training device on CPAP utilisation data in patients with moderate to severe OSA.

In 2021, I published two pre-print papers, a retrospective analysis of minute-ventilation variability during cardiopulmonary exercise in disordered breathing patients. Secondly, in a service evaluation examining if exhaled breath temperature predicts maximal exercise performance in a population of males with COPD. I performed the data collection, analysis, and manuscript preparation for these publications.

Career Trajectory

I have clear clinical academic aspirations which I have explored with my line manager, General Manager and the Director of Clinical Research who is responsible for nursing and allied professional clinical academic career development at St Barts Health. We are proposing to create a unique clinical academic Consultant Clinical Scientist in Sleep Medicine post for my post-doctoral role, using the PhD fellowship to include the clinical, academic and leadership training required for that role.

My ultimate ambition is to become a Professor in Sleep Medicine and part of the next generation of research leaders.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
In November 2022 I applied for funding with Barts Charity, unfortunately, I was unsuccessful but, in the process, received valuable positive feedback. *“The panel felt that the proposal was of great clinical importance. The NICE guidelines specifically called for more*

research in this area so the potential for immediate impact from this grant is significant. Previous studies evaluating MAD treatment for patients with OSA have usually excluded those with severe disease. Due to lack of evidence, current guidelines do not recommend that MAD be used to treat patients with severe OSA. The proposed study will fill this gap in our knowledge by determining the role of this treatment in patients with severe disease".

"The proposal is outstanding, exceptionally well designed. Objective measures will be used to assess the use and efficacy of the two interventions. Subjective outcomes will be used to assess functional outcomes. Structured qualitative interviews will be administered to evaluate participant perspectives and preferences. A cost-utility analysis will be performed to compare the two interventions. The use of the TheraMon sensor for tracking objectively the adherence to MAD treatment is innovative. Relatively few research studies have employed this novel technology. The proposed approaches are all appropriate and feasible. The applicant details an excellent training plan. The considerable number of sleep center patients available for participation should allow successful recruitment of the required volunteers.

However, given the emphasis on health economics, mean disease alleviation, QoL, cost-effectiveness the panel felt that the applicant would have better success and academic mentorship from a public health specialist, rather than two clinical academics. There is also a significant qualitative element to this research and neither supervisor has this expertise".

To address these comments, I redesigned my supervisory team. I have successfully secured the support and guidance of Dr. Fiorella Colonio Salazar, an expert in health-related quality-of-life research who will act as a secondary supervisor. This collaboration represents a significant opportunity to enhance the quality of my academic output and highlights my ongoing commitment to this fellowship.

I re-engaged with RDS-London to explore various PPI strategies. During our discussions, we focused on participant engagement and recruitment strategies. I had the privilege of networking with a former member of the NIHR Advocate Scheme. This encounter not only provided me a further review and input in the application but also a unique opportunity to gain insights into the lived experiences of individuals who have actively participated in research projects.

Engaging in conversations with this experienced advocate broadened my perspective and deepened my understanding of the diverse range of considerations within the research landscape.

Lastly, I was incredibly fortunate to establish a new network connection, with Professor Robert Adams. He is an internationally recognised researcher in respiratory care and leads a program of sleep health research at the University of Adelaide (UoA) Australia. Prof Adams has promised to act as an international collaborator and steering group member. We are keen to establish stronger links between UoA and QMUL, and I have been offered an international placement.

4. Plain English Summary of Research

Research Aims

To explore the effect of mandibular advancement therapy on the physical, mental, social and emotional health and wellbeing of people living with the severe obstructive sleep apnoea (OSA).

Research Background

OSA affects breathing during sleep, in which there is repeated partial or complete obstruction of their airway, resulting in dangerously low levels of oxygen in the body.

This obstruction to breathing can significantly impact on the person's quality of sleep, with sleep deprivation resulting in an increased risk of road traffic accidents, memory loss and mood disturbance. The lack of oxygen has been shown to increase the risk of developing high blood pressure, heart attack and stroke.

The accepted treatment for OSA is continuous positive airway pressure (CPAP), in which the airway is prevented from collapsing by a machine which blows air through a tight-fitting face mask. However, many people struggle to successfully use this machine for several reasons, among which the mask can feel claustrophobic the machine is noisy, and the air pressure is uncomfortable to the eyes and causes bloating of the stomach.

A recognised alternative to CPAP is a mandibular advancement device (MAD), these resemble a gum shield and when worn at night, they hold the lower jaw and tongue forward, preventing the airway from collapsing. Importantly, patient's report greater preference for an MAD in place of CPAP. The current research is much needed, given the low patient acceptance of CPAP, leaving them vulnerable to a significant healthcare burden and the high patient expressed preference for MAD treatment. As yet, the benefits of MAD treatment in patients with severe OSA need to be demonstrated in order to offer these as an alternative option for care.

Design and Methods

I plan to conduct a clinical trial involving patients with severe OSA, designed to compare CPAP and MAD treatments looking at changes in their breathing, blood pressure and heart function. Patients will also complete questionnaires designed to look at their general health and mood status. One of these questionnaires is concerned with helping to calculate the costs of providing CPAP and MAD treatments and therefore help to determine their value-for-money. In total, the study will require three visits to the hospital over a six-month period.

I also plan to invite these patients to take part in interviews, designed to explore their views on the two treatments and help identify what they feel are key factors for us to consider in helping to improve the treatments.

Public involvement and engagement

Prior to committing to this application, I undertook clinical observation in OSA-specialist clinics, and I spoke with patients with suspected-OSA. I engaged with the British Society of Dental Sleep Medicine and patient charities, OSA Alliance and the OSA Partnership Group.

Patients articulated the need to improve their quality of life as they experienced challenges such as fatigue, tiredness, loss of concentration and irritability. Equally, patients described the need to identify symptoms and to improve individual treatment modalities to successfully improving functional aspects of living with OSA. Because of

this engagement I was able to form a group of four patient / expert users who were prepared to support me with setting up this project.

Dissemination

The product of this research will be shared by the appropriate format in scientific conferences and publications in peer-reviewed journals. It will be shared with stakeholder networks and support groups, both online and via engagement activities to exchange knowledge and raise awareness of my work. Potentially my research could provide proven benefit to patients with this severely debilitating breathing disorder and cost-effectiveness in its approach to care.

5. Scientific Abstract

Background

Obstructive Sleep Apnoea (OSA) is a prevalent respiratory disorder in the UK, affecting up to 10% of middle-aged adults, with increasing prevalence due to the rising levels of obesity and aging population. This condition is associated with significant health implications and has direct costs estimated to be over £55 million annually to the National Health Service. OSA is characterised by the repetitive narrowing or closure of the upper airway during sleep, leading to hypopnoea or apnoea, resulting in hypoxia and consequent awakening from sleep. Patients with OSA often experience excessive daytime sleepiness, poor concentration, personality changes, irritability, and are at increased risk of heart disease and stroke in the longer term.

The standard treatment for moderate to severe OSA is continuous positive airway pressure (CPAP) via a face mask, which prevents the upper airway from collapsing during sleep. However, some patients may be unable to tolerate this treatment. In such cases, mandibular advancement devices (MAD) may be an alternative therapy; MAD acts to hold the lower jaw and tongue forward, thus maintaining airway patency, making more space to breathe.

Despite the increasing use of MAD as an alternative therapy, limited evidence exists regarding its effectiveness in severe OSA patients. The 2021 NICE Guidelines for OSA recommend further research into the impact and effectiveness of MAD in these patients. Therefore, this fellowship aims to investigate the feasibility of using MAD as an alternative therapy in severe OSA patients.

Research question

"*What is the clinical and cost effectiveness of mandibular advancement devices in managing severe OSA compared to standard CPAP therapy over a six month period?"*

Objectives

1. To evaluate if MAD treatment achieves an equivalent level of short term 6-month effectiveness compared to CPAP therapy, as measured by the mean disease alleviation (MDA).
2. To investigate if patients using a MAD will have a better quality of life than those using CPAP?
3. To determine if MAD will provide cost effective therapy compared to CPAP.

Methods

A prospective mixed method feasibility study will be conducted. A total of 52 consecutive patients with a recent diagnosis of severe OSA, meeting the selection criteria will be allocated to the intervention (n=26) and standard care (n=26) groups. Comparing both CPAP and MAD treatments and clinical outcomes like blood pressure, cardiac and respiratory function. Participant interviews and questionnaires will be utilised to fully identify and measure the impact and treatment on quality of life. Finally, a cost analysis will be completed to understand the cost implications of both treatments.

Timelines for delivery

Participant recruitment will take place at Barts Health Sleep Centre. The estimated recruitment timeline is approximately 40 weeks. The trial will be conducted in a 36-month period with assessments at baseline, 3 and 6 month post intervention follow up.

Anticipated impact and dissemination

The anticipated impact of this research is to provide robust evidence on the clinical and cost effectiveness of MAD in treating severe OSA, with results disseminated through peer-reviewed publications and scientific conferences.

6. Detailed Research Plan

What is the problem being addressed?

Current estimates suggests that nearly one billion adults aged 30–69 years worldwide could have obstructive sleep apnoea (OSA), and the number of people with moderate to severe OSA, for which treatment is generally recommended, is estimated to be almost 425 million[1] with prevalence set to rise, due to the increasing age and levels of obesity in the population[2]. Direct costs to the NHS currently are estimated to be over £55 million annually[3]. OSA involves repetitive narrowing or closure of the upper airway during sleep, which results in reduced (hypopnoea) or a complete cessation (apnoea) of breathing. This abnormal breathing causes oxygen levels to drop and leads to repeated awakening from sleep.

In turn, this sleep disturbance results in excessive daytime sleepiness, poor concentration, personality changes and irritability. In the long term, there is an increased risk of heart disease and stroke in affected individuals. OSA is a chronic condition that generally requires life-long treatment. The standard treatment for people with moderate to severe OSA is continuous positive airway pressure (CPAP), where air is blown into the upper airway during sleep, via either a nasal or face mask, to prevent it from collapsing. However, some patients are unable to tolerate this treatment and along with mild to moderate OSA, can be treated with mandibular advancement devices (MAD) which hold the lower jaw and tongue forward, making more space to breathe.

Given the health implications and the limited evidence, the 2021 NICE Guidelines for OSA recommend research should be conducted into the impact and effectiveness of using MAD in severe OSA [4].

Why is this research important in terms of improving the health and wellbeing of the public and to service users?

The consequences of untreated OSA include excessive daytime sleepiness leading to increased risk of road traffic accidents, impaired cognitive function, reduced quality of life (QoL) and increased vascular risk [5,6]. Patients with severe OSA , have a 3-fold increased risk of developing hypertension over four years, independent of other risk factors [7]. It has been predicted that over 10 years, 23% of patients with severe OSA will have a stroke, and 14% will have a myocardial infarction[7]. It is estimated that up to 1% of middle-aged males in the UK suffer from severe OSA this represents a considerable healthcare burden. Effective treatment would substantially reduce health care costs and improve the QoL of both patients and their families [5].

Given, the reported poor tolerance or simple refusal towards the gold standard therapy, CPAP, there is an ethical imperative to offer alternative treatments [8–10],[11,12]. The recently published NICE guideline recognises the role of MAD in OSA but highlights a need to understand their effectiveness as an alternative treatment modality, particularly when acknowledging patient preference for it over CPAP[13].

This sentiment was demonstrated by patient comments in some of my early PPI work:

“That face mask wouldn’t do much for my sex life”.

“The CPAP looks so bulky, and I would be afraid of getting caught up in the pipe and strangle myself”.

Patient-centred care recognises the value of addressing the unique needs and preferences of individuals. This research is therefore important to identify the health and/or wellbeing benefits of MAD therapy to this vulnerable group.

How does the existing literature support this proposal?

NICE and the James Lind Alliance both acknowledge that patients with severe OSA deserve an informed choice to individually tailored alternative treatments, such as oral appliances[14-15].

A key issue with prescribing CPAP or MAD therapy is that they are both entirely patient-dependent treatments, requiring consistent use, per night, seven days a week to be effective. Despite being a highly effective treatment

many do not tolerate CPAP, and the low compliance undermines the effectiveness of therapy [16]. There is no fixed definition of therapy compliance and adherence rates range from 40% to 85% [17]. In the US, compliance has been arbitrarily defined as device usage for more than 4 hours per night for more than 70% of nights[18,19].

Bespoke MAD, are mouthpieces worn at night, permitting gradual advancement of the lower jaw and can be an effective alternative treatment option in mild-moderate OSA and those patients unable to tolerate CPAP [20]. Importantly, a series of Cochrane reviews have confirmed a significant patient preference for MAD over CPAP, due to negative side effects and noted that long-term comparisons of treatment effectiveness and particularly objective assessment of MAD compliance in severe OSA are required [21–23].

Previous trials focussed on short-term (1-3 months) treatment periods or mild OSA patients only [24–26]. The currently accepted measure of the severity of OSA is the number of apnoea/hypopnoea episodes per hour (apnoea-hypopnoea index; AHI), measured by an overnight sleep study. The accepted cut-offs for severity are mild (AHI 5–15), moderate (AHI 16–30), and severe (AHI >30 per hour). Whilst AHI measures sleep apnoea severity over the night, it does not consider compliance when comparing treatments. With CPAP therapy, the device monitors and records compliance, in contrast, MAD rely on subjective reporting to estimate treatment use. This makes it difficult to obtain a true and accurate assessment of their effectiveness and long-term usage.

More recently, the safety and feasibility of objective measurements of MAD use in patients who were unable to tolerate CPAP treatment has been demonstrated in a pilot study[27–29]. Whilst CPAP therapy remains more effective in controlling OSA in the short term, given the lifelong nature of the treatment, this effectiveness is limited by its lower compliance in the long term. In these instances, patients may fail to comply with treatment at all or have sub-optimal compliance, not achieving the minimum four hours per night or not using it every night[30].

Furthermore, published pilot data support the long-term use of MAD therapy in patients unable to tolerate CPAP[27]. However, further research in severe OSA, which includes an objective measure of compliance and evaluates the long-term comparative effectiveness of CPAP and MAD, is required[31,32,21]. Thus, the importance of being able to accurately record compliance objectively becomes apparent. To resolve this, a novel method of assessing outcome will be used, the mean disease alleviation).

There is growing evidence demonstrating the utility and validity of mean disease alleviation index, a combination of physiological outcome measures, the AHI, with a record of compliance/usage of the treatment and thus, provides a better marker of treatment effectiveness than conventional endpoints. Specifically, mean disease alleviation is equal to the surface area of the plot for which the length is given by the adjusted compliance (use/total sleep time), and the height is given by the therapeutic efficacy (AHI baseline minus AHI with mean disease alleviation applied, expressed in percentage)[33–46,28,29,47]. (Fig 1). Mean disease alleviation is important in assessing the scale of treatment required to control symptoms.

NICE concluded, *"There is evidence for the effectiveness of oral devices in mild and moderate OSA but not in severe OSA, nor in people who are CPAP intolerant*". This research proposal addresses this short coming by trialling an existing intervention combined with physiological and QoL outcomes.

In terms of the relevance to the NHS, NICE also state, *"a clear recommendation will offer clinicians clearer guidance on the use of CPAP or MAD in people with severe OSA and would be cost effective as the appropriate most effective treatment would be selected initially, minimising failure rate from less effective treatment and potentially needing to try both therapies"* [14].

There is a clear necessity to address this knowledge gap, using robust research. It is my intention to provide evidence to patients and healthcare communities of alternative treatment modalities for the treatment of severe OSA.

What is the research question / aims and objectives?

Research Question

"*What is the short term clinical and cost-effectiveness of mandibular advancement devices in managing severe OSA compared to standard CPAP therapy?"*

Aims

To investigate the treatment and cost-effectiveness of mandibular advancement devices on patients with severe OSA.

Objectives

1. To evaluate if MAD treatment for OSA achieves an equivalent level of short term (i.e. 6-month) clinical effectiveness compared to CPAP therapy, as measured by the mean disease alleviation.
2. To investigate if patients using a MAD will have a better quality of life than those using CPAP?
3. To determine if MAD is potentially more cost-effective therapy compared to CPAP.

Primary Objective: Pilot randomised clinical trial with patient interviews

This study will be reported in accordance to the SPIRIT Statement[48].

Hypothesis

1. Null Hypothesis: There will be no statistically significant difference in mean disease alleviation at 6 months between MAD and CPAP therapy.
2. Null Hypothesis: MAD use will not achieve AHI <5 per hour at 6 months compared to CPAP therapy.
3. Null Hypothesis: The use of MAD will not yield a 50% or greater reduction in AHI at 6-months compared to baseline.

Study Design

A triangulated convergent mixed methods study, employing a randomised clinical trial, (parallel intervention) and qualitative research (face-to-face semi-structured interviews).

Participant Inclusion Criteria

Adults who can provide consent with a recent diagnosis of severe OSA and sufficient healthy teeth to allow satisfactory MAD fabrication.

Participant Exclusion Criteria

Any coexistent sleep disorder, drug treatment or unstable cardiovascular disease, considered likely to have significant impact on symptoms or the effectiveness of the intervention. Any form of seizure history or psychological disorders.

Recruitment Strategy

Recruitment will take place at the Institute of Dentistry, Queen Mary University of London (QMUL) and Department of Respiratory and Sleep Medicine, St Bartholomew's Hospital. The latter is a recognised referral sleep centre and receives between 80 and 200 new patient referrals per month, performing between 30 to 55 sleep studies per week, it is estimated that 25% of these result in a diagnosis of severe OSA. Estimated recruitment timeline is approximately 40 weeks. I will perform all the recruitment tasks: eligibility, screening, consenting.

Recruitment Risks

Pilot Study - Potential risks are non-completion of outcomes, lower recruitment and service changes due to pandemics. Loss or damage to equipment resulting in a replacement that is beyond budgetary scope.

Patient Centred Outcomes - non-completion of questionnaires or engagement during interviews.

Cost Utility Analysis - No expected risks

Planned Study Period

The study is to be conducted in a 36-month period with a 3 and 6 month post intervention follow up.

Sample Calculation

I contacted an independent consultant statistician (Ron Horne - rhstatistics.com) for the purposes of discussing the statistical analysis plan. I stressed; (1) this is a pilot study (2) research design (3) published literature demonstrating the effect size[50] (4) potential dropout rate. G*Power was used to calculate sample size.

The sample size calculation assumed:

1) Two groups

2) Three repeated measures per arm.

3) Type I error: $\alpha=0.05$

4) Power (1- Type II error): $1-\beta=0.8$.

5) A small effect size of 0.30.

Under these assumptions, the sample size estimation is 20 participants, with a Degrees of Freedom of 2. It is expected that 30% of participants will drop out and will be adjusted accordingly. Under these assumptions, the sample size estimation is **52 participants in total, 26 per arm**. (Fig 1)

Statistical Analysis

Descriptive statistics for continuous variables will be presented as means and standard deviations for normal distributions, or medians and interquartile ranges for skewed distributions. Categorical variables expressed in terms of proportions.

To compare differences between baseline and follow-up variables within groups, the paired t-test or Wilcoxon signed-rank test will be used for skewed distributions. The differences between the intervention and standard care treatment groups compared by a repeated measures ANOVA with Bonferroni. If the data have a skewed distribution, a Friedman test can be performed.

Randomisation

Patients will be randomly assigned to intervention (MAD) or standard care (CPAP) group, using baseline AHI (>30/hour) as a single stratification factor, with random block sizes of between 2 and 6 using Graphpad Prism. A total of 52 consecutive patients with a recent diagnosis of severe OSA, meeting the selection criteria, will be allocated to the intervention (n=26) or standard care (n=26) groups. In this non-blinded study bias will be prevented by (1) standardised protocols and level of care (2) structured data monitoring (3) regular supervisory meetings.

Intervention Group

Sleepwell MAD is custom-made (S4S Ltd) device, which conforms to medical device regulation (CD93/42/EEC & SI No. 618: 2002). A thermo-microsensor (TheraMon, AT), will be embedded in the MAD to record compliance. Data will be subsequently transmitted and analysed, using radiofrequency identification technology (TheraMon, v2.1) (Fig 1).

Standard Care Group

The Air Sense 10 Autoset CPAP (ResMed, UK) device will be provided and conforms to medical device regulation (IEC60601-1-2:2014). Patients will be offered a change of mask and/or chin straps or humidifier to improve use if deemed necessary (Fig 1).

Patient Safety Considerations

Transient side effects are commonly reported in the initial period of MAD therapy including; excessive salivation, tooth discomfort, bite disturbance[32]. Symptoms mostly resolve within days to weeks with regular use and appropriate adjustment of the device [51–53].

In comparison, side effects of CPAP use include airway dryness, nasal congestion, facial skin irritation, eye irritation. Some users may experience only mild side effects, while others may experience more severe side effects that make it difficult to tolerate CPAP therapy[54].

Secondary Objective - Patient Centred Outcomes

Introduction

This element has been developed in accordance with COREQ checklist[55], therefore promoting explicit reporting strategies for qualitative studies. The purpose of convergent methodology is to obtain emergent but complementary data, to better understand the research problem[56]. I acknowledge challenges of this design, issues of sample size, the need to merge text and numerical outputs and a need to explain the divergence when comparing results.

Stage 1: Qualitative Element

Objective

The use of semi-structured interviews to explore the patients' views and experiences on their QoL following use of

MAD and CPAP.

Study Design

A convergent parallel mixed-method study, conducted in a 36-month period. Interviews will be performed at the 3- and 6-month post intervention follow up.

Study participants:

Inclusion and Exclusion Criteria

The inclusion and exclusion criterion are detailed above.

Patient sample / recruitment process

Recruiting participants from the control study. Criterion-based purposive sampling based on specific criteria will be used to select participants; a sample matrix of age, gender, severity of OSA, and intervention compliance will be utilised in this study.

Given the PhD timeframe and based on previous qualitative research in the dental field, it is anticipated that up to 24 participants (12 participants per arm) will be sufficient to explore the depth and breath of the phenomenon[57], however the number of participants required for this element will be determined by reaching data saturation, when no new information was heard or seen in at least three successive interviews[58].

Instrument

The use of a topic guide enables flexibility in exploring emerging themes and issues, while maintaining consistency and comparability across participants. A pre-piloted topic guide will be utilised and developed based on current literature, professional knowledge, and around the research objectives. The topic guide will be piloted during the initial fieldwork stage; the first 3-4 interviews. This data will be excluded from the final data analysis. Any necessary alterations to the topic guide and/or any other aspect of the interview will then be made prior to commencing the full data collection process.

The key areas covered in the topic guide include the multifaceted influence of treatment on QoL. Specifically, (1) the impact of functional issues and symptoms, (2) self-esteem and psychosocial well-being, and (3) suggestions for enhancing treatment effectiveness. Furthermore, participants will be encouraged to raise any additional aspects relevant to their treatment that were not explicitly covered in the topic guide. Throughout the interview process, the topic guide will undergo regular refinement to accommodate the evolving nature of the research, facilitating the collection of novel data and enabling further exploration of topics previously discussed in earlier interviews.

Data collection:

One-to-one interviews will be conducted by myself with participants in a non-clinical and quiet meeting room, within St Barts Hospital, ensuring a secure and confidential environment. Participants' personal information will be securely stored and accessed only by authorised personnel. The interviews will be recorded using a digital voice recorder, ensuring that accurate and complete data capture is achieved. The recorded files will be securely transferred to an external company [TakeNote Ltd] for verbatim transcription, adhering to data protection protocols. All transcripts will be thoroughly reviewed and checked for accuracy before being used for data analysis. In addition to the audio recordings, I will use a notepad to take field notes and record observations. The notes will focus on capturing non-verbal cues, participants' demeanour, and my personal insights and reflections, providing valuable contextual information for data analysis.

Data Analysis

To uphold anonymity and confidentiality, participants' identities will be kept strictly confidential, participant codes will be used in any notes. To ensure data rigor and trustworthiness, framework methodology[57,59] will be used to analyse the qualitative data, which involves: 1) Familiarising with contents of interview transcripts 2) Labelling topics identified from each interview as 'codes', 3) Grouping codes sharing similar characteristics into 'themes', 4) Construction of a framework table, with columns of codes and themes, and rows of participants, and 5) Mapping data from each interview into the framework. Data will be managed by NVivo software.

Reflexivity

I will aim to conduct this element with reflexivity, a continual process of engaging with and articulating the place of the researcher within the context of the research. This may involve at times challenging and articulating social and cultural influences and dynamics that affect this context[60]. As a qualified sleep specialist, I have knowledge of the mechanisms of sleep and its disorders. I sometimes encounter patients experiencing fatigue, anxiety and emotional destress. These experiences may inadvertently result in unconscious bias, a failure to acknowledge the patient experience, "personal interactions, organisational culture, patient and family perceptions, and continuum of care" [61].

To minimise interpretation bias and strengthen the validity of findings, the interview transcripts will be analysed independently by myself and the secondary supervisor, verify the emerging categories from the thematic framework. Any disagreement will be resolved by discussion and consensus. Furthermore, the study participants will be new to the research team and me; thus, enhancing the reliability of findings.

Outcome

An assessment of patient-reported outcomes; QoL, functional status, or satisfaction with the intervention. These measures capture the patients' perspective on how the intervention impacts their daily life and well-being.

Stage 2: Quantitative Element

Questionnaires capture crucial insights into treatment efficacy, safety, and patient-reported outcomes. Different questionnaires are to be used to assess specific domains, such as symptom severity, functional status, and treatment satisfaction. I acknowledge the issue of frequency and the potential risk of participant burden and attrition. Achieving a careful equilibrium between collection and burden is paramount to ensure the acquisition of accurate and reliable responses.

Objective

To investigate if patients using a MAD will have a better quality of life than those using CPAP using sleep related questionnaires?

Study Design

Conducted in the same 36-month period, validated questionnaires will be completed at baseline visit and the 3 and 6 month post intervention follow up.

Inclusion and Exclusion Criteria

As stated in the parallel control trial.

Sample size

The same 52 subjects from the parallel study (n=26 standard care, n=26 intervention) will be used for the sleep related questionnaires.

Data collection/Recruitment process:

Questionnaire completion will be conducted on a 1:1 basis in a separate quiet room, within the St Barts Hospital. Questionnaire completion will part of all visits and last approximately 30 minutes [62]

Instrument

Study participants will complete the following validated questionnaires:

Functional Outcomes of Sleep Questionnaire

Assesses the impact of excessive daytime sleepiness on social interactions, vigilance, intimacy.[63,64]

Short-Form Health Survey (SF-36)

QoL was assessed using the SF-36: physical functioning, role, bodily pain, health perceptions, vitality, and mental health [65].

Sleep Apnoea Quality of Life Index (SAQLI)

The SAQLI measures the impact of OSA on daily functioning, symptoms, emotional well-being. [66]

Sleepiness

Daytime sleepiness was assessed using the Epworth Sleepiness Scale. Participants rated their likelihood of falling asleep in different situations. [67] [68,69]

Data analysis

Descriptive statistics, including measures of central tendency and variability, summarise the scores obtained from the questionnaires.

Subscale analysis allows for a more focused assessment of specific domains or components within the questionnaires.

Comparative analysis compares scores between different groups or time points to determine significance.

Internal consistency and reliability analysis assesses the reliability of questionnaire items.

Correlation analysis examines relationships between questionnaire scores and other variables of interest.

Outcome

A record of valuable insights into the personal experiences of individuals dealing with sleep-related issues. By comparing the scores before and after interventions, I can evaluate the effectiveness of these interventions in improving the specific areas assessed by each questionnaire.

Triangulation process

To achieve triangulation, qualitative data will be thematically analysed to identify recurring patterns, participants' narratives, and observations, providing in-depth understanding of their experiences and perceptions. Quantitative data will undergo statistical analysis to generate numerical data, revealing trends and statistical significance, therefore offering a broader understanding and facilitate generalisation to a wider population. After separate analysis, the data will be compared and merged to identify convergence or divergence. Examining how findings align, complement, or provide contrasting perspectives will enhance the overall validity and reliability of the study [70–73].

Tertiary Objective - Cost Utility Analysis

Introduction

Cost-utility analysis will be used to determine if differences in expected costs between CPAP and MAD can be justified in terms of changes in expected health effects[74]. Health effects will be expressed in terms of *Quality-Adjusted Life Years* (QALYs). The use of QALYs is justified, measuring health effects, both quantity (life years) and quality of life and combined into a single output with minimal burden on patients. Despite the limitations of QALYs, cost-effectiveness studies that incorporate QALYs provide a useful indication of whether new interventions provide good value for money[75–77]

Objective

To determine if MAD will provide cost effective therapy compared to CPAP.

Study Design

The economic tool recommended by NICE is the *EQ-5D-5L*, to calculate QALYs when comparing technology [74]. The EQ-5D-5L is an internationally recognised instrument of five dimensions: mobility, self-care, activities, pain and depression[78]. The output of EQ-5D-5L will be “cross walked” to allow comparison with known standard values as shown by Van Hout *et al* and recommended by NICE [79–81]. Economic evaluations will adhere to CHEERS checklist [82].

Participants will complete the *EQ-5D-5L* at the 6-month post intervention visit only.

Inclusion/Exclusion Criteria

Previous stated

Recruitment

The same 52 subjects from the parallel study (n=26 standard care, n=26 intervention) will be used for the cost-utility analysis.

Outcome

Following standard economic conventions, the output of QALY's will be used to calculate the *Incremental Cost Effectiveness Ratio* (ICER) and will be presented as the *Cost Effective Acceptability Curve*.

Dissemination, Outputs, Anticipated Impact

What do you intend to produce from your research?

This research is expected to produce three peer-reviewed publications:

1. A study protocol submitted to Trials BMC. 2) A pilot study submitted to the *European Respiratory Journal* and 3) Health attributed cost analysis submitted to the *Health Economics Review Journal*.

2. The output of this PhD will be disseminated by posters and presentations at these conferences: *Association for Respiratory Technology & Physiology* (2024), *British Sleep Society* (2024 & 2028) and *European Sleep & Research* (2028).

This PhD has an anticipated impact as an outcome measure for research, particularly in clinical trials and clinical practice. With the emerging realisation from eminent bodies such as NICE, JLA and the European Respiratory Society, recognising there is insufficient research and thus is a global need to explore the use of MAD in severe OSA. This pilot study serves as a solid foundation to my potential post-doctoral research, namely, a multi-centred randomised controlled trial hopefully with the support of an advanced NIHR Fellowship.

How will you inform and engage service users and the wider population about your work?

I recognise the critical importance of informing and engaging service users, services, and the wider population about my work. I will be supported and guided in the PPI efforts by the NIHR Barts Biomedical Research Centre, a partnership between Barts NHS Trust and QMUL.

My strategy for stakeholder engagement is:

Service Users:

Regular communication: I will establish ongoing communication channels with patient advisory groups or support organisations (OSA Charity). This will enable me to provide updates on the progress of the research and share information and input. Ten meetings (6-monthly) will help to facilitate and coordinate PPIAG involvement during the fellowship. In addition, I will continue engaging with my PPI via emails and/or telephone calls to discuss research progress and troubleshoot problems (e.g. challenges with recruitment or retention).

Plain language summaries: To ensure that service users can easily understand and access the key outcomes and implications of the study.

Healthcare Services:

Stakeholder workshops: I will organise workshops involving HCPs, and policymakers to present and discuss the research findings. This will encourage dialogue, collaboration, and the exchange of knowledge.

Professional conferences: I will actively participate in relevant conferences to share my work with a wider audience of HCPs. These events provide a platform for networking, collaboration, and dissemination.

Wider Population:

Public lectures and talks: I will engage with the wider population by delivering public lectures and talks on topics related to my research. Specifically, I will use *Sleep Awareness Week, National Stop Snoring Week* and *World Sleep Day* as a driver to raise awareness and educate the public about the implications of my work. Furthermore, I will use *Healthcare Science Week* to raise awareness of my research with other HCPs. I plan to collaborate with NIHR CRN North Thames during *International Clinical Trials Day.*

Online platforms: I will leverage online platforms, such as social media, research blogs, and academic websites, to share key messages related to my research. This will allow for wider accessibility and engagement with a diverse audience. Currently avenues being considered are a LinkedIn Research Page, a YouTube Channel, Facebook and Instagram Live events.

How will your outputs enter our health and social care system or society as a whole?

This PhD will produce an output of three peer-reviewed publications and four conference engagements. Furthermore, it is my hope that this work contributes to improvements in healthcare policy and practice. For instance, the outcome and publication of the pilot trial and trial design serve as a solid foundation to my potential post-doctoral research, namely, a multi-centred randomised controlled trial hopefully with the support of a NIHR Advanced Fellowship.

The publication regards the cost effectiveness of MADs, highlights the health economics of such devices so that policymakers make informed decisions about the allocation of resources, leading to more equitable and accessible healthcare for OSA patients.

What further funding or support will be required if this research is successful?

None required

What are the possible barriers for further research, development, adoption and implementation?

Despite the use of MAD for the treatment of OSA being a promising avenue for development several barriers that must be overcome. A primary challenge is the lack of standardisation in the design and fabrication of MADs. The diversity of MADs available with unique design features could hinder meaningful comparison across studies and impede the identification of the most effective devices[83] .

Another barrier limiting MAD adoption and implementation is cost. Financial burden can make it difficult for many patients with OSA to access these devices, particularly if they are not covered by insurance[84]. Furthermore, inadequate awareness and education about the potential benefits of MADs among healthcare professionals and patients can result in underutilisation of the devices and slow adoption rates[85].

What do you think the impact of your research will be and for whom?

To enhance severe OSA management, the potential of MAD for improved QoL and reduced comorbidity risks through strengthened treatment guidelines. Moreover, healthcare systems could see cost savings from ameliorated OSA management, reducing the burden on healthcare infrastructures. The impact could be substantial, the prospect of a paradigm-shift towards elevated standards of OSA care.

Project Management

a. Joint Research Management Office and the NIHR's Coordinated System for gaining NHS Permission for research management, financial aspects, and governance.

b. QMUL's PhD programme includes progression milestones at set intervals over the 5-year period.

c. Monthly meetings with my supervisors to review progression and plan future work.

d. Multidisciplinary steering committee will meet at 6-monthly intervals to evaluate the study progress.

Ethics

The research will adhere to QMUL's good research practice, NIHR Research Governance Framework and Good Clinical Practice.

All research governance and approval requirements, including obtaining Research Ethics Committee and Health Research Authority approval via the Integrated Research Application System, will be obtained. Ethical requirements (e.g., consent, withdrawal, confidentiality, data security, GDPR) will be adhered to. I will comply with all local sponsorship requirements. All specific materials (e.g., protocol, PIS) will be submitted to QMUL/Barts R&D, for review prior any formal submission to REC.

Interviews and questionnaires may lead to patients disclosing sensitive information, strong emotions, or distressing memories, the Trust's psychology/patient wellbeing service are aware and supportive of my plans, and I am currently establishing a patient contact pathway. Any Adverse Events will be reported, and safety actions taken. I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce CO_2 emissions generated by this research.

Success Criteria

Achieved recruitment within specified timeframe.

Meeting PhD milestones.

Peer-reviewed publications and conferences.

A clinical academic career.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

PPI activities align with the GRIPP2 Checklist[29], promoting transparent and high-quality PPI reporting in research. I strive to boost accessibility and inclusivity, involving a diverse range of individuals.

I engaged with the British Society of Dental Sleep Medicine and patient charities, OSA Alliance. Initial input involved informal discussions with patients and family members. This provided insights into their perspectives, highlighting the need to enhance their life amidst challenges like fatigue, concentration loss, and irritability. Patients also expressed the necessity to identify symptoms and improve treatment modalities for better living with OSA. Through RDS-funded efforts, I actively sought the opinions, expectations, and concerns of service users and the public, using open-ended questions to capture their insights and difficulties.

I established a Patient and Public Involvement Advisory Group (PPIAG) of four patient/expert users from diverse and minority groups; one participant is BAME one identifies LGBTQ. There is a wide range of educational and socioeconomic backgrounds also represented.

One member of the group provided feedback on the proposal:

"*One comment I would like to make, if I may, is how very worthwhile this proposed study would be. This study would add to our knowledge in many areas which is a good thing. It would also make excellent use of new technology (i.e. the MAD chip) which is a wonderful resource for patients.*"

Their contributions encompassed:

Recommending questionnaires in both paper and electronic formats.

Reviewing and shaping the lay statement.

Devoting time to understanding study design and measurement endpoints.

Iterative feedback integration ensured the PPIAG's perspectives were embedded into the proposal. Feedback analysis led to adjustments after each round, including changes to the plain English summary to translate scientific terms.

In conclusion, public involvement in proposal development, particularly through the PPIAG, underscores our commitment to inclusivity, transparency, and responsiveness. This active engagement enhanced our proposal, addressing immediate audience needs while aligning with broader societal values. The multifaceted involvement of the PPIAG demonstrates its importance in anchoring research and policy initiatives in the perspectives and needs of those they intend to benefit.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

<u>Ethical Review and Consent Processes:</u>

The PPIAG will ensure research remains respectful of participant's rights, privacy, and cultural values. Their insights will leave an indelible mark on the construction of consent forms, data management protocols, and other ethical facets of the study, bolstering their acceptability and comprehensibility.

<u>Participant Recruitment:</u>

Potentially identify participants, share information, and address queries, fostering trust for a diverse participant pool.

<u>Acknowledgement:</u>

Recognised in publications and presentations, highlighting their vital role and fostering pride.

Training

The training provided to the PPIAG will equip members with the knowledge, skills, and confidence to effectively engage in research. All members will complete a NIHR GCP certificate. Further training will be provided QMUL's Public Engagement service, it is free and predominantly online.

Research Literacy: This empowers them to understand the research process and effectively contribute to discussions and decision-making.

Critical Appraisal Skills: PPIAG members will develop the ability to critically assess research proposals, methodologies, and findings. This empowers them to provide constructive feedback and enhance research quality.

Data Interpretation: Basic training in interpreting research data enables PPIAG members to provide valuable insights during data analysis and discussions. This enhances their contribution to the research process.

Feedback and Reflection: Providing constructive feedback and engaging in reflective practices helps PPIAG members continually improve their contributions.

The combination of these training areas will empower PPIAG members to engage effectively, offer informed insights, and contribute meaningfully to this PhD.

We plan to have 10 meetings for the duration of this 5-year fellowship.

All PPI activities have been diligently costed and the reimbursement for participants according to the INVOLVE guidelines and reviewed by RDS-London.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
This section is not applicable to my application.

8. Training & Development and Research Support

Proposed training and development programme

Following the Vitae Researcher Development Framework, my proposed training and development programme is based on a gap analysis designed to provide me with the skills necessary to complete this fellowship and support my continued academic, clinical and leadership development.

As a PhD student at the Institute of Dentistry, I will have access to the wide range of training provided by *QMUL Doctoral College* which will include core PhD training modules as well as additional short courses and workshops aimed at my holistic development as a future clinical academic leader.

PhD Specific

Essentials Of Clinical Trials Course builds on my existing clinical trials experience with advanced clinical trial design, protocol assessment and validation.

Trial Management and Governance will ensure good Trial Master Documentation and GCP compliance.

Qualitative Research Training focussing on interviewing. **Analysing Qualitative Interviews,** the principles of qualitative interview data analysis. **NVivo Course** to perform thematic analysis of the interview data.

Clinical Development

Diploma in Sleep Medicine, this course will provide a superb platform for my theoretical learning and development.

Clinical Assessment, to develop knowledge, skills, and understanding that underpin clinical assessment in the context of differential diagnosis. Ultimately this will allow me to operate within a sphere of personal competence and level of authority suitable for progression to Consultant Clinical Scientist.

Independent Prescribers', I will be able to analyse, synthesise, evaluate, and apply knowledge of appropriate respiratory-specific therapeutic interventions.

Leadership Development

Certificate in Leadership explores the key concepts of leadership and change management. I will learn where it is appropriate to apply theory and research relating to management, leadership and team effectiveness.

Certificate in Teaching in Higher Education, I will develop skills and strategies in teaching, learning and assessment in face to face opportunities.

Overseas Visit

I have arranged an overseas visit, I will spend time at Professor Robert Adams department, Flinders University / The University of Adelaide Australia. Being guided by Prof Adams and other research colleagues (Prof Danny Eckert, MAS Expert and Prof Ching Li Chai-Coetzer, an experienced sleep physician with expertise in clinical trials). During my placement, I intend to immerse myself in research work, recruit volunteers, perform the outcome measures and analyse and interpret the results with the potential to publish some of this work.

The training plan detailed above offers not only a breadth of academic learning and attainment but also dovetails directly into my clinical workload. With each new topic learnt, it can be applied directly to my sleep service meaning immediate patient benefit. As a commitment to my continual career progression, we are submitting for additional funding to be used in the creation of a new post; *Trainee Consultant Clinical Scientist - Head of Sleep Physiology (8A)*, this provides a financial incentive for the duration of the fellowship. Upon completion of the PhD Fellowship, this post will be promoted to a Consultant Clinical Scientist (8B) post in preparation for my post-doctoral ambitions.

Details of the practice that will be undertaken over the course of the fellowship.

My role as a Trainee Consultant Clinical Scientist necessitates advanced skills in the assessment and management of a variety of sleep disorders. To maintain these skills as well as ensure continuing competence, I will continue to lead St Barts sleep services.

From early 2024, the North East London Respiratory Clinical Network will establish six new individual diagnostic hubs to cover activity e.g. general asthma and COPD, integrated care, TB and Sleep Medicine. These clinics will require advanced new diagnostics and treatment pathways and be led predominantly by Consultant Nurses and Consultant AHPs. I will lead the Sleep Medicine element of these for the Trust.

The training plan is designed to provide balance and development throughout the entirety of the 5-year fellowship (GANTT Chart).

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Prof Ama Johal

Institution	Queen Mary University London / Barts Health NHS Trust

Position	Professor - Lead for orthodontics

Description of support provided
Prof. Johal (AJ) is a renowned expert in the application of mandibular advancement devices for the treatment of obstructive sleep apnoea. He was a NICE committee member for the development of the first UK guideline on sleep-related breathing disorders [2021]. AJ's impressive track record includes successfully supervising 8 PhD and 15 Taught Doctorate candidates to completion. All of these PhD students have been completed within the required timeline and all, but one have progressed to Academic careers. Noteworthy instances comprise one individual's appointment as a Professor at QMUL and their more recent appointment at Dublin University. Moreover, a former PhD student, who embarked on an academic journey through the Academic Clinical Lectureship [via HEE], has currently advanced to a senior Lecturer position. AJ has been fortunate with all his students and managed to mentor the majority beyond their PhD. Additionally, AJ has maintained an active publication record from 2021 to 2023, with a specific focus on mixed-method research and the application of mandibular advancement devices in the treatment of sleep apnoea. AJ's mentorship will be comprehensive and personalised equipping me with the skills needed to succeed in my future academic career. The clinical research will take place in the Department of Orthodontics at the Royal London Dental Hospital ranked 1st in London and 15th worldwide for dentistry and its management of sleep-related breathing disorders was recognised specifically in the Care Quality Commission report of the Dental Hospital.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Dr Fiorella Colonio-Salazar

Institution	Queen Mary University London

Position	Secondary Supervisor

Description of support provided
Dr Fiorella Colonio-Salazar (FCS) is a Lecturer in Orthodontics and Dental Public Health and has demonstrated a deep understanding of the intersection between dentistry and public health. FCS has recent (2021-2023) publications in mixed-method research techniques and the utilisation of mandibular advancement devices for sleep apnoea treatment. FCS's expertise spans various aspects of dental public health, encompassing areas such as epidemiology, health promotion, and oral healthcare policy development. Her research contributions have advanced academic understanding but also had a real-world impact on improving oral health practices and policies. As a mentor, FCS offers me support and guidance in preparation for my academic career and will be a key factor in the success of the qualitative and interview elements of my proposed PhD.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Prof. Julie Sanders

Institution	Queen Mary University London / Barts Health NHS Trust

Position	Professor / Clinical Director Of Research

Description of support provided
Prof. Sanders (JS) has a key role that focuses on: a) Developing research and clinical academic career opportunities for nurses and allied health professionals (NAHPs) b) Serving as the director of clinical research and providing clinical academic support to all pre-doctoral, doctoral, and post-doctoral nursing and allied professional staff at St Bartholomew's Hospital. Over the past 6 years, this role has resulted in 31 successful competitive fellowship awards. JS has offered invaluable clinical academic support and mentorship to four staff members who completed their PhDs during this time, in addition to seven HEE/NIHR internships, three HEE/NIHR pre-doctoral fellowships, one HEE/NIHR pre-doctoral silver award, and one HEE/NIHR post-doctoral fellowship. Currently, JS supervises three PhD students and provides post-doctoral mentorship to a further three individuals. Most recently, JS has been my academic mentor for the NIHR/HEE Pre-Doctoral Bridging Programme. We have conducted weekly teleconferences and monthly face-to-face supervisions. Throughout the PhD program, we will continue to meet regularly, evolving to a minimum of monthly meetings (or as needed) to ensure the successful completion of the PhD. Overall, JS possesses the clinical academic expertise necessary for this PhD, and SBH/QMUL offers a dynamic and supportive NAHP clinical academic research environment. This environment provides me with a strong support network and setting in which to pursue my PhD and subsequent post-doctoral clinical academic work.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Prina Ruparalia

Institution	Barts NHS Trust

Position	Consultant in Sleep & Ventilation / Clinical Director

Description of support provided
Dr. Prina Ruparalia (PR) leads Sleep Medicine and Ventilation at St Barts NHS Trust. PR has acted as a secondary supervisor to one MD student. PR has established a specialised service for non-invasive ventilation patients at Barts Health NHS Trust, where monthly multidisciplinary clinics are conducted in collaboration with respiratory physicians and other specialist professionals. St Barts boasts the largest ventilation service in the United Kingdom. With its strong sleep foundation, this service provides comprehensive care for individuals with OSA. PR has been a valuable resource in shaping the research design and facilitating Patient and Public Involvement engagement in my proposal. PR involvement in my clinical development will provide me with mentorship, specialised training and networking opportunities.

Collaborations
I have established national and international clinical and academic collaborations/links that will be central to this fellowship and my future clinical academic career. **METHODOLOGICAL COLLABORATIONS/ADVISORY SUPPORT** Health economics and statistical collaborations/advisory support: Barts Health NHS Trust and Queen Mary University of London, London, UK. **COLLABORATIONS/ADVISORY SUPPORT/MENTORSHIP** Dr Aditi Desai, President of the British Society of Dental Sleep Medicine (BSDSM), dental sleep expert and dental surgeon. Has agreed to act as an ad hoc mentor and a conduit to professional networking. Mr Matthew Norie, Technical Director S4S (UK) Limited, will support my learning of mandibular advancement devices. As indicated in the application a visit to the manufacturing site is planned. Mrs Karen Schofield, Key Account Manager, ResMed UK and Mr Pedro Nogueira, Application Specialist, NOX Medical, Reykjavík Iceland will support where appropriate around CPAP therapies and sleep diagnostic testing respectively. Flinders University / The University of Adelaide Through my supervisor Professor Julie Sanders, collaborative links with international academics have been made. Professor Robert Adams is an internationally recognised researcher in respiratory care and leads a program of acute care and primary health services research at Flinders University. As indicated in my proposal, Professor

Adams has agreed to a study visit to his academic group to a) learn more about obstructive sleep apnoea in Australia and the current research activity b) establish links with multidisciplinary sleep experts, c) examine opportunities for further collaboration research. He has agreed to be part of the PhD steering committee.

Professor Sheree Smith is the Director of the Lung, Sleep and Heart Health Research Network. Professor Smith is a member of the SPHERE Respiratory Sleep, Environmental and Occupational Health Research Clinical Academic Group. Professor Smith is committed to developing and conducting high-quality innovative research that improves patient outcomes through developing, testing and implementing sustainable models of care and cost-effective interventions. She also holds a postgraduate certificate in Health Economics. She has agreed to be part of the PhD steering committee.

Host Organisation Support Statement
The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

As the sleep and ventilation lead and joint clinical director at St Bartholomew's and Royal London Hospital, Bart's Health NHS Trust, I am pleased to write in support of Andrew Eaton for completing this Research Fellowship.

Mr. Eaton currently works as a clinical scientist at Barts and the London Hospitals, where he holds the position of Lead for sleep physiology. His extensive clinical experience in sleep medicine, coupled with a keen interest in advancing research on mandibular advancement splints for the treatment of severe obstructive sleep apnoea, aligns seamlessly with his clinical expertise. Whilst he currently works as a full-time clinical scientist, should he be successful in this award he will be given dedicated protected research time of 60% whole time equivalent and will continue to provide 40% clinical activity within the department. The remaining clinical time will be back provided by appointing additional clinical staff to ensure that Andrew's research time is protected. Clinical work will include review of patients requiring sleep diagnostics and will align and facilitate recruitment of patients to this research study.

We actively encourage and support our researchers to have dedicated time and develop new professional skills which will enhance their future clinical practice.

I can confirm the department has the capacity and capability for this type of study, and that the study is consistent with the departmental research strategy. Our respiratory department has dedicated space and facilities and works within a wide multi-disciplinary team. Supervision for this project will be provided by Professor Ama Johal, Dr Fiorella Colonio-Salazae and Professor Julie Sanders. Dr Prina Ruparelia (sleep and ventilation lead St Bartholomew's Hospital) will act as a practice supervisor. The CPAP service cares for approximately 8000 patients with sleep apnoea. We receive 30 to 50 referrals per week. Our catchment area for sleep apnoea referrals is predominantly from North East London.

The respiratory department is committed to providing early researchers' support by providing dedicated time for research.

In a collaborative effort with Prof. Johal and the Institute of Dentistry, the Department of Sleep and Ventilation is exceptionally well-suited to endorse and facilitate Mr. Eaton's proposed doctoral research, which builds on the existing care pathway for the management of obstructive sleep apnoea. As a large tertiary centre for CPAP and NIV patients, the department boasts a robust support system, encompassing a dedicated team of medical, nursing, and physiotherapy professionals.

Andrew has shown commitment to undertaking research in the field, which makes him an ideal candidate for this Fellowship. This collaboration holds great promise in contributing valuable insights to the field of sleep medicine and ultimately improving the quality of care for patients with obstructive sleep apnoea. Andrew has undertaken a pre-doctoral fellowship in order to progress his research proposal.

The trust has a large diverse staff and patient population and the leadership team and staff promote an inclusive

environment. We are committed to supporting access to research opportunities to our patient population and early career researchers are vital in promoting this. The Heart and Thorax centre board overviews and supports early career researchers.

Andrew's expertise in this field means that he is ideally suited to engage in this area of research. I think he is an excellent applicant for this Research Fellowship. Our patient population will enable recruitment to answer this research question. Support will be provided by regular research meetings with his supervisors to review progress towards a PhD.

In terms of future career we would aim to provide career progression to develop a consultant clinical scientist post with the remit of promoting education, academic and clinical excellence. quality of care for patients with obstructive sleep apnoea.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Andrew Mark Eaton

Employing organisation	Barts Health NHS Trust

Grade	8A

Starting point on scale	1 - Entry

Increment Date	01/08/2024

Justification of Costs
Mr Eaton will work 0.60 FTE as an 8A Trainee Consultant Clinical Scientist on the project for 5 years; he will recruit patients to the study, collect data, analyse the results and disseminate the findings.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	51485.08
Geographical Weighting (£)	7746
Other allowances (£)	0.00
Superannuation (£)	12793.914
Nat. Insurance (£)	7771
Current Annual Costs (£)	**79795.99**

2

Staff type	Shared Staff

Name	NOT YET NAMED

Employing organisation	Queen Mary University of London

Role on this research	Statistician

Grade	7PRO

Starting point on scale	049

Increment Date	01/08/2029

Justification of Costs
The QMUL Pragmatic Clinical Unit in conjunction with the Joint Research Management Office have confirmed that statistical and health economist resource will be allocated to this project once funding has been confirmed. The statistician will be contributing specialist advice in the areas of statistical analysis. This will be applied to ensure outcomes are well defined, appropriate statistical tests are chosen and performed. Utilising a statistician also allows for independence of the analysis of results. Statistical support will be provided in year 5 of the fellowship

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	69275.92
Geographical Weighting (£)	
Other allowances (£)	
Superannuation (£)	15215.59355
Nat. Insurance (£)	9327.26708333333
Current Annual Costs (£)	**93818.78**

3

Staff type	Shared Staff

Name	NOT YET NAMED

Employing organisation	Queen Mary University of London

Role on this research	Health Economist

Grade	7PRO

Starting point on scale	049

Increment Date	01/08/2029

Justification of Costs
The QMUL Pragmatic Clinical Unit in conjunction with the Joint Research Management Office have confirmed that statistical and health economist resource will be allocated to this project once funding has been confirmed.

The health economist shall provide expert input pertaining to cost analysis. They will guarantee the accurate completion of the EQ-5D-5L, the proper calculation of Quality-Adjusted Life Years, and the presentation of the Incremental Cost-Effectiveness Ratio (ICER) within the framework of the Cost-Effective Acceptability Curve.

Furthermore, the inclusion of a health economist statistician enables an impartial evaluation of the outcomes.

Health economist support will be provided in year 5 of the fellowship

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	69275.92
Geographical Weighting (£)	0.00
Other allowances (£)	0.00
Superannuation (£)	15215.5935
Nat. Insurance (£)	9327.26708333333
Current Annual Costs (£)	**93818.78**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mr Andrew Eaton
Type of cost: NHS
% Full-time on this research: 60.0
Total months on this research: 60.0
Justification of Cost: Mr Eaton will work 0.60 FTE as an 8A Trainee Consultant Clinical Scientist on the project for 5 years; he will recruit patients to the study, collect data, analyse the results and disseminate the findings.

	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Total Cost (£)**
Basic Salary	51486	57349	57349	57349	57349	280882
National Insurance	7771	8680	8680	8680	8680	42491
Superannuation	12794	14061	14061	14061	14297	69274
Geographical weighting / Other allowances (if applicable)	7746	7746	7746	7746	7746	38730

Staff Type: Other
Other Type: Shared Staff
Name: **Not Yet Named**
Type of cost: HEI
% Full-time on this research: 2.0
Total months on this research: 1.0
Justification of Cost: It is estimated that the project will require 4 days of statistical support. This will take place in year five and costings have been estimated accordingly.

	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Total Cost (£)**
Basic Salary	0	0	0	0	1519	1519

Staff Type: Other
Other Type: Shared Staff
Name: **Not Yet Named**
Type of cost: HEI
% Full-time on this research: 2.0
Total months on this research: 1.0
Justification of Cost: It is estimated that the project will require 4 days of economist support. This will take place in year five and costings have been estimated accordingly.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Basic Salary	0	0	0	0	1519	1519

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	2430	2430
Total NHS	431377	0	0	431377
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: Registration fee
Justification of Cost: These costs are for: Association for Respiratory Technology & Physiology (2024) - £800 British Sleep Society (2025 & 2028) - £800 European Sleep & Research (2028) - £800

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	800	400	0	0	1200	2400

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Travel cost to conference
Justification of Cost: These costs are for: Association for Respiratory Technology & Physiology (2023) - Return Off Peak Rail Ticket with Disable Persons Rail Card – Total £65.60 British Sleep Society (2024 & 2027) -– Return Off Peak Rail Ticket with Disable Persons Rail Card – Total £131.20 European Sleep & Research (2027) - Return Economy Plus Airplane Ticket – Total £375

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	66	0	0	0	375	441

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Return travel to conferences
Justification of Cost: Justification of Cost: These journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. These costs are for: Association for Respiratory Technology & Physiology (2024) - Return Off Peak Rail Ticket with Disable Persons Rail Card – Total £65.60 British Sleep Society (2025 & 2028) -– Return Off Peak Rail Ticket with Disable Persons Rail Card – Total £131.20 European Sleep & Research (2028) - Return Economy Plus Airplane Ticket – Total £375

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	66	132	0	0	507	705

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: Accommodation to attend conferences
Justification of Cost: These accommodation costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. Accommodation will be arranged and approved via the Agiito Connect as per Barts Trust Policy. These costs Association for Respiratory Technology & Physiology (2024 - 3 Days – £200/day – Total £600 British Sleep Society (2025 & 2028), 800 x 2 Brighton - 6 days – £250/day – Total £1500 European Sleep & Research (2028) - 4 days - £350/day – Total £1400

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	600	750	0	0	2150	3500

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Subsistence (food & beverage) costs to attend conferences
Justification of Cost: These costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. These costs are for Association for Respiratory Technology & Physiology (2024). 3 days – Breakfast/Lunch/Dinner – Total £120 British Sleep Society (2025 & 2028), 6 days – Breakfast/Lunch/Dinner – Total £240 European Sleep & Research (2028), 4 days - Breakfast/Lunch/Dinner – Total £180

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	120	120	0	0	300	540

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Return travel to training course venue
Justification of Cost: These journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. Oxford University – 2 days - Off Peak Rail Ticket with Disable Persons Rail Card at £97.70 return – Total £195.40 S4S Dental Laboratory Ltd S4S Dental Laboratory Ltd – 1 day - Off Peak Rail Ticket with Disable Persons Rail Card – Total £210.75 Swansea University – 8 days – Off Peak Rail Ticket with Disable Persons Rail Card at £108 return – Total £864 Cambridge University – 16 days – Drive Personal Car, 88 mile return trip at 56 pence per mile. – Total £788.48 University West London – 12 days – Zone 1 to 7 TFL Travel Card at £14.30 – Total £171.60

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	407	0	864	961	0	2232

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Subsistence (food & beverage) costs to attend training courses
Justification of Cost: These journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. These costs are for Swansea University – 12 days – Breakfast/Lunch/Dinner – Total £480 Oxford University - 7 days – Breakfast/Lunch/Dinner – Total £280 Cambridge University – 16 days – Lunch – Total £160 University College London – 10 days – Lunch – Total £100 Brunel University - 3 days – Lunch – Total £30 Institute for Clinical Research - 3 days– Lunch – Total £30 S4S Dental Laboratory Ltd – 1 day – Lunch -Total £10

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	420	30	480	160	0	1090

Type of cost: NHS
Description: Subsistence
Conference: False

Item Description: Accommodation to attend training courses
Justification of Cost: These accommodation journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and the Worldwide subsistence rates (06-Apr-2019) for international conferences. Additionally, these costs are estimated considering their respective location for 2023. These costs Swansea University – 12 days at £48/night – Total £576 Oxford University - 7 days at £120/night – Total 840

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	576	0	840	0	0	1416

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Accommodation for International Training Visit
Justification of Cost: Accommodation for International Training Visit 14 nights of hotel accommodation is required and is within reasonable walking distance to the University Of Adelaide, this was chosen for two reasons (1) no additional cost incurred from commuting and (2) as a disabled adult with mobility issues a long commute may be physically arduous or painful. The hotel cost is £1498. This journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk#countries-a-to-c. If this cost exceeds the NIHR limit any remainder will be met by the Host Trust.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	1498	0	0	0	0	1498

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Subsistence for International Training Visit Adalaide, Australia
Justification of Cost: It is assumed that Breakfast is included in the accommodation fee. I will only need lunch and an evening meal during my 14-night stay away from home. This totals $1,862 Australian Dollars or £969 Pounds. These costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001) and https://www.gov.uk/guidance/expenses-rates-for-employees-travelling-outside-the-uk#countries-a-to-c. If this cost exceeds the NIHR limit any remainder will be met by the Host Trust

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	969	0	0	0	0	969

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Travel, Subsistence and Conference Fees Total	5522	1432	2184	1121	4532	14791

Equipment

Type of cost: NHS
VAT relief to be applied: False
Description: Encrypted laptop LG gram 16 16Z90R-K.AA78A1 16" Laptop - Intel® Core™ i7, 1 TB SSD, Black
Price excluding VAT £: 1000
Expected life time: 60
Justification of Cost: A laptop is required to access QMUL/Barts network where study data will be stored. This laptop is required to conduct literature searches statistical analysis; and research report writing. the interviews and audio files will also be interrogated and analysed

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	1000	0	0	0	0	1000

Type of cost: NHS
VAT relief to be applied: False
Description: Encrypted USB memory-sticks
Price excluding VAT £: 130

Expected life time: 36
Justification of Cost: Justification of Cost: 2 x Trust approved encrypted USB memory-sticks (£65 each) will enable me to securely save and transport data between study sites and to my research collaborators. The cost for these reliable items has been calculated after comparing characteristics and prices between competitive sellers.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	130	0	0	0	0	130

Type of cost: NHS
VAT relief to be applied: False
Description: Samsung Galaxy Tablet
Price excluding VAT £: 200
Expected life time: 60
Justification of Cost: Justification of Cost: This tablet is required to enable participants to complete questionnaires electronically before their routine outpatient clinic. This option was recommended by my PPI advisory group. The cost for this reliable item has been calculated after comparing characteristics and prices between competitive sellers

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	200	0	0	0	0	200

Type of cost: NHS
VAT relief to be applied: False
Description: Dictaphones
Price excluding VAT £: 128
Expected life time: 60
Justification of Cost: Description: Price excluding VAT £: 170 Expected life time: 60 Justification of Cost: 2 x Olympus WS-853 digital voice recorders to record semi-structured face-to-face interviews (£64 each, ex-VAT). The cost for these reliable items has been calculated after comparing characteristics and prices between competitive sellers.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	128	0	0	0	0	128

Type of cost: NHS
VAT relief to be applied: False
Description: Lockable briefcase for the safe transfer of confidential materials such as note books and the digital voice recorder
Price excluding VAT £: 42
Expected life time: 60
Justification of Cost: Justification of Cost: Lockable briefcase for the safe transfer of confidential materials such as laptop, USB memory sticks, notebooks, tablet and the digital voice recorder

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	42	0	0	0	0	42

Type of cost: NHS
VAT relief to be applied: False
Description: Lockable notebooks
Price excluding VAT £: 17
Expected life time: 12
Justification of Cost: Justification of Cost: Lockable notebooks (x4) will be required for the safe documentation of observational notes and notes from the semi structured interviews.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	17	17	17	17	0	68

Type of cost: NHS
VAT relief to be applied: False
Description: Theramon reading hardware and software licence
Price excluding VAT £: 1629
Expected life time: 60
Justification of Cost: Theramon reading hardware and software licences required for downloading the embedded Theramon radio frequency sensor in the mandibular advancement device.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	1629	0	0	0	0	1629

Type of cost: NHS
VAT relief to be applied: False
Description: Theramon USB-BT-PEN-Reader
Price excluding VAT £: 578
Expected life time: 60
Justification of Cost: Theramon USB-BT-PEN-Reader required for downloading the embedded Theramon radio frequency sensor in the mandibular advancement device.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Year 4 (exc VAT)	Year 5 (exc VAT)	Total Cost (£) (exc VAT)
	578	0	0	0	0	578

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Equipment Total	3724	17	17	17	0	3775

Consumables

Description: General Stationary
Type of cost: NHS
Please describe fully: General Stationary
Justification of Cost: Justification of Cost: General stationary costs for the project have been estimated at £10/month. These include paper (for meeting agendas, draft documents etc)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	120	120	120	120	120	600

Description: General Printing and Photocopying
Type of cost: NHS
Please describe fully: General Printing and Photocopying
Justification of Cost: Justification of Cost: It is required to print patient information sheets (PIS), informed consent forms (ICF), study questionnaires, screening/baseline/follow-ups forms (CRF). Additionally, printing is required for dissemination leaflets and PPI collaboration work. £100 per year (10p per sheet x1000 estimated pages per year).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	100	100	100	100	100	500

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Consumables Total	220	220	220	220	220	1100

Patient and Public Involvement

Description: Donation to a group/network
Type of cost: NHS
Please describe fully: Donation to a group/network
Justification of Cost: This cost includes five donations (£100 each year) to recognise (OSA Group) support for this project. This cost has been calculated based on the INVOLVE guidelines and reviewed by my PPIAG and RDS-advisers

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	100	100	100	100	100	500

Description: Catering for biannually meetings with PPIAG
Type of cost: NHS
Please describe fully: Catering for biannually meetings with PPIAG
Justification of Cost: This cost includes funding for catering service to provide refreshments and snack/food (£50 per meeting) in the ten meetings (twice a year) with my PPIAG. This cost has been calculated based on the INVOLVE guidelines and reviewed by my PPIAG and RDS-advisers.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	100	100	100	100	100	500

Description: Payment and recognition for PPI time - Reviewing documents
Type of cost: NHS
Please describe fully: Payment and recognition for PPI time - Reviewing documents
Justification of Cost: This cost includes payment for time to recognise activity involvement of my PPI advisers during the study, considering £100 per PPI adviser for five scenarios (set-up and ethics, item generation, item selection, analysis, and dissemination). My PPIAG is constituted by four advisers. This cost has been calculated based on the INVOLVE guidelines and reviewed by my PPIAG and RDS-advisers

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	400	400	400	400	400	2000

Description: Payment and recognition for PPI time - Meetings
Type of cost: NHS
Please describe fully: Payment and recognition for PPI time - Meetings
Justification of Cost: This cost includes payment for time to recognise activity involvement of my PPIAG during the study, considering the attendance of face-to-face meetings (£75). My PPIAG is constituted by four advisers. This cost includes ten biannual Steering Committee and PPIAG meetings ((4 PPI advisers meeting twice a year for 5 years)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	600	600	600	600	600	3000

Description: Travel expenses for meetings.
Type of cost: NHS
Please describe fully: Travel expenses for meetings.
Justification of Cost: Using the St Barts Trust travelling booking system (Agiito Connect) it is estimated that than average travelling cost of £50 using public transport This cost has been calculated based on the INVOLVE guidelines and reviewed by RDS-advisers.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	400	400	400	400	400	2000

Description: Disclosure and Barring Service check enhanced
Type of cost: NHS
Please describe fully: Disclosure and Barring Service check enhanced
Justification of Cost: This cost includes payment for Disclosure and Barring Service (DBS) checks for each PPI adviser as recommended by INVOLVE guidelines. The cost for each DBS enhanced check is £40 (£40 x 4 = £160).

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	160	0	0	0	0	160

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Patient and Public Involvement Total	1760	1600	1600	1600	1600	8160

Other Direct Costs

Description: Open access publications
Type of cost: NHS
Please describe fully: Open access publications
Justification of Cost: These costs include open access publications during the fellowship. This research is expected to produce three peer-reviewed publications in the following journals: 1) Trials BMC ≈ £ £2390.00 + VAT, 2) European Respiratory Journal (≈£1,755+VAT), 3) Health Economics Review (≈£1790+VAT). These open-access publication costs are based on each website considering appropriate APC rates as of 2023. If costs exceed NIHR limit remaining cost will be met by NHS Host via CPD funds.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	2868	2106	2148	7122

Description: Creating of scientific posters for conferences and presentations.
Type of cost: NHS
Please describe fully: Creating of scientific posters for conferences and presentations.
Justification of Cost: This cost is to allow results to be presented as posters to clinicians, researchers, policymakers, patients and the public. It is estimated that six A0 posters will be required over the duration of the fellowship. TheCopyShop at QMUL Campus (www.copyshop.qmul.ac.uk), at £30 each per poster has been included.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	60	60	60	180

Description: Archiving costs
Type of cost: NHS
Please describe fully: Archiving costs
Justification of Cost: This cost is to allow the study archiving process based on the appropriate standard operating procedure (SOP) from the Joint Research Management Office (QMUL/Barts) £574

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	0	0	574	574

Description: Interview transcription costs
Type of cost: NHS
Please describe fully: Interview transcription costs
Justification of Cost: TakeNote Ltd is a London-based transcription service (https://takenotetyping.com/transcription-services), certified by the ICO (company ref 00041352899) and certified in ISO 27001 (internationally recognised best practice framework for an information security management system) and 9001 (certified quality management system). This service is used widely for academic research transcription. It is a cloud-based (Amazon AWS UK/Ireland) (no information is ever downloaded) and all audio and transcription files are automatically deleted 3 and 12 months after project end, respectively. Therefore, using this service is robust from an information governance perspective. Costs are based on £64.60 per hour of recorded audio (if 1-3 speakers) using the 7 day turnaround service (longest possible time available). This research includes 24 interviews (maximum of 90minutes per interview), total cost will be £1550.40

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	388	388	388	388	1552

Description: Graphpad Prism licence – Statistical Data Analysis software (Analysis & Randomisation)
Type of cost: NHS
Please describe fully: Graphpad Prism licence – Statistical Data Analysis software (Analysis & Randomisation)
Justification of Cost: I require an easy-to-use but powerful statistical analysis for the duration of the fellowship. The student-rated option is £100 per year.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	100	100	100	100	100	500

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Other Direct Costs Total	100	488	3416	2654	3270	9928

Training And Development

Type of cost: NHS
Description: Applicant PhD Tuition Fees
Item Description: Application PhD Tuition Fees
Justification of Cost: This cost includes 2024-2025 tuition fees for PhD at Institute of Dentistry Queen Mary University of London

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	4712	4712	4712	4712	4712	23560

Type of cost: NHS
Description: Short Courses
Item Description: Training and development courses
Justification of Cost: This costs include the fees to attend the training and development courses stated in my training and development plan: Year 1: PG Diploma in Sleep Medicine (£7,315), Qualitative Research Method (£1,670), Qualitative Research interview Skills(£700) ,Essentials of Clinical Trials (£1000), Skills & Competencies of a Clinical Trial Manager (£1000). Year 2: Health Economics (£1000), Medical Statistics Modules (£ 2500) Year 3: Clinical Assessment (£1500) Year 4 Prescribing Course (£ 4000), PG Leadership course (£ 5,000.00) Year 5 PG Teaching course (£ 2,000),

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	11685	3500	1500	9000	2000	27685

Type of cost: NHS
Description: Overseas Research Visits
Item Description: 2-week international training visit at the University Of Adelaide
Justification of Cost: I have a 2-week international training visit at the University Of Adelaide, under the guidance of Prof Robert Adams (Respiratory and Sleep) and Professor Danny Eckert. leading authority on mandibular advancement devices. A return flight from London to Adelaide, Australia at a cost of £2044. These journey costs are calculated according to the current Barts Trust Employee Expenses Policy (COR/POL/229/2018-001). Kindly be aware that, as a registered disabled adult, due to exceptional mitigating circumstances and the duration of this flight, I have reserved premium economy seats.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	2044	0	0	0	0	2044

Type of cost: NHS
Description: Short Courses
Item Description: Expert Sleep Technologist examination
Justification of Cost: Upon successful culmination of the Postgraduate Diploma in Sleep Medicine, I intend to undertake the Expert Sleep Technologist examination. This accomplishment will elevate my status as one of the most highly qualified sleep medicine practitioners in the country. This will hopefully coincide with the ESRS conference although it is possible to complete on line. The exam fee is € 600 - £523.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
	0	0	523	0	0	523

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Training And Development Total	18441	8212	6735	13712	6712	53812

Total Direct Costs

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total Direct Costs	109564	99805	102008	107160	106836	525373

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total Indirect Costs	0	0	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	2430	0	2430
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	522943	0	522943
Total Research Costs Requested From Funder	525373	0	525373

Summary

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Salaries	**79797**	**87836**	**87836**	**87836**	**90502**	**433807**
Travel-subsistence-and-conference-fees	**5522**	**1432**	**2184**	**1121**	**4532**	**14791**
Equipment	**3724**	**17**	**17**	**17**	**0**	**3775**
Consumables	**220**	**220**	**220**	**220**	**220**	**1100**
Patient-and-Public-Involvement	**1760**	**1600**	**1600**	**1600**	**1600**	**8160**
Other-Direct-Costs	**100**	**488**	**3416**	**2654**	**3270**	**9928**
Training And Development	**18441**	**8212**	**6735**	**13712**	**6712**	**53812**
HEI-Indirect-costs	**0**	**0**	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**	**0**	**0**

Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**	**0**	**0**
Total	**109564**	**99805**	**102008**	**107160**	**106836**	**525373**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Support Costs	2288	0	0	0	0	2288.0

Justification:
NHS Support Costs: Please refer to the authorised SoECAT form for a detailed line-by-line explanation of study costings and savings.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Support Costs Total	2288	0	0	0	0	2288

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Treatment Costs	65321	0	0	0	0	65321.0

Justification:
NHS Treatment Costs: This is the cost of providing the Standard Care Arm (CPAP) of the parallel study. Please refer to the authorised SoECAT form for a detailed line-by-line explanation of study costings and savings. It is assumed that NHS Treatment Cost is the same as Usual Treatment Costs

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
NHS Treatment Costs Total	65321	0	0	0	0	65321

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Novel Care Arm - MAD	58093	0	0	0	0	58093.0

Justification:
Novel Care Arm - MAD: This is the cost of providing the Novel Care Arm (MAD) of the parallel study. Please refer to the authorised SoECAT form for a detailed line-by-line explanation of study costings and savings.

	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Usual Treatment Costs Total	58093	0	0	0	0	58093

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	7228	0	0	0	0	7228

	Total (£)
NHS Support Costs requested from networks	2288
NHS Excess Treatment costs/savings	7228
TOTAL NHS SUPPORT AND TREATMENT COSTS	9516

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	525373
Total NHS (Support and Excess Treatment) Costs	9516
Total Funding Required	534889

10. Uploads

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - Figures .pdf) is included as an appendix within this file.

(Research Timetable - Research Timetable.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_328478_SoECAT_Mandibular advancement devices in severe sleep apnoea.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Dr
Forename(s)	Ama
Surname	Johal
Position	

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Fiorella
Surname	Colonio-Salazar
Position	

Supervisor Participant 2

Title	Professor
Forename(s)	Julie
Surname	Sanders
Position	Director of Clinical Research

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	Prina
Surname	Ruparelia
Position	Consultant

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Dr
Forename(s)	Ama
Surname	Johal
Position	

Head of Department or Senior Manager 2

Title	Dr
Forename(s)	Prina
Surname	Ruparelia
Position	Consultant

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
Forename(s)	Coleen
Surname	Colechin
Position	Operations Manager

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

No conflicts of interest to declare.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References.pdf
2. Figures .pdf
3. Research Timetable.pdf
4. Funder Export_328478_SoECAT_Mandibular advancement devices in severe sleep apnoea.xlsx

Appendix #1: References.pdf

References

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Appendix #2: Figures .pdf

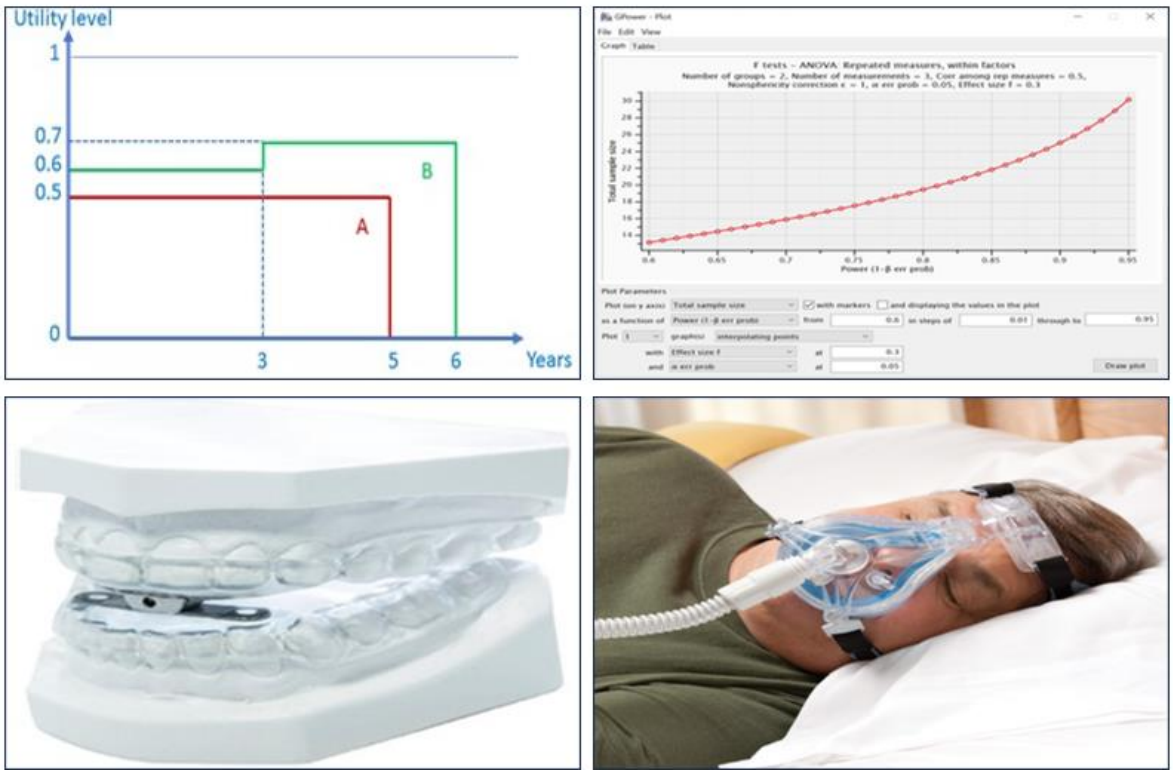

Fig1: Mean Disease Alleviation, Sample Calculation, Mandibular Advancement Device, Full Face CPAP

Appendix #3: Research Timetable.pdf

Year	2024				2025				2026				2027				2028			
Quarter	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project management and governance																				
Project steering group (2 per year)																				
Supervisory meetings (2 per quarter)																				
Patient and public involvement																				
Project milestones																				
Study recruitment																				
Data collection																				
Data analysis																				
Preparation of papers/reports/ summaries of results																				
Training																				
International Visit																				
Diploma in sleep medicine																				
Essentials of clinical trials																				
Clinical assessment																				
Clinical trial management																				
Medical statistics modules																				
GCP refresher																				
Certificate in leadership course																				
QMUL teaching course																				
Health economics																				
Independent Prescribers'																				
Conferences/Publication/Output																				
Association for Respiratory Technology & Physiology																				
British Sleep Society																				
European Sleep Research Society																				
Study protocol publication																				
Mix method study publication																				
Health cost analysis publication																				
Doctoral thesis																				

Appendix #4: Funder Export_328478_SoECAT_Mandibular advancement devices in severe sleep apnoea.xlsx